AIL7110: Introduction to Augmented Reality and Virtual Reality

Assignment - 02

WorldForge: Robotics Lab

Development of a Virtual Environment Using Unity

Abstract

This project presents WorldForge: Robotics Lab, an interactive 3D virtual environment developed using Unity. It incorporates realistic materials, textures, lighting, and first- and third-person navigation, along with interactive elements such as a robotic arm, conveyor belt, and physics-based objects. The project demonstrates fundamental concepts of virtual environment design and interaction, creating an immersive and functional robotics laboratory.

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1 Objective and Environment Overview

The objective of this assignment was to build a 3D virtual environment in Unity and use it to practise the basic concepts of virtual world creation: modelling and placing objects, designing materials and lighting, navigating with a camera, and interacting with the scene.

I built **WorldForge Robotics Lab**, a single-room robotics laboratory that the user explores in first person. The room is 20 m × 14 m with a 4 m ceiling. It has a red-brick back wall, concrete side walls and a continuous glass wall on the north side, which looks out over open green ground under a daylight sky. The room is organised into five zones:

- **Robot cell** (centre-left): an orange 4-axis robot arm on a pedestal, standing on a diamond-plate floor marked with yellow-and-black hazard stripes, with two small tables it moves a crate between.
- **Conveyor line** (along the brick wall): an 8 m belt that carries wooden crates between an input and an output housing.
- **Workstations** (east wall): two desks with monitors showing a code editor and a telemetry dashboard.
- **Workbench** (centre): a wooden bench holding a crate, a rubber ball and a toolbox the user can pick up.
- **Storage** (west wall): a metal shelf with boxes and bins, and a whiteboard showing the robot arm's equations.

2 3D Objects: Creation and Placement

All objects were built inside Unity from primitive shapes (cubes, cylinders, spheres and quads), scaled and combined into larger objects. No external 3D models were imported. Every object is built at real-world scale (a desk is 0.75 m high and the eye height is 1.62 m), so the space reads correctly in first person. The final scene contains about 200 objects, 162 of which are rendered meshes.

Object	How it is built	Placement
Room shell	Floor, ceiling, roof and three solid walls; the glass wall is a curb, 7 metal mullions, 8 glass panes and a top beam	Encloses the whole lab
Robot arm	Pedestal, rotating base, upper arm, forearm, wrist and two-finger gripper, each part attached to the previous joint	Centre of the robot cell
Pick/place tables	Steel top, column and foot with a yellow target mark	Two sides of the robot cell
Conveyor	Belt, side rails, legs, two end housings with light strips, five crates	Along the brick wall
Workstations (×2)	Desk, monitor with screen, keyboard and chair	East wall
Workbench	Top, legs and lower shelf, with a crate, ball and toolbox on top	Centre of the room
Shelf	Four uprights, four boards and nine boxes and bins	West wall
Details	Ceiling light panels, whiteboard, three signs, two potted plants, light switch	Walls and corners
Outside	Large flat grass ground plane	Around the building

Table 1: Main 3D objects in the environment.

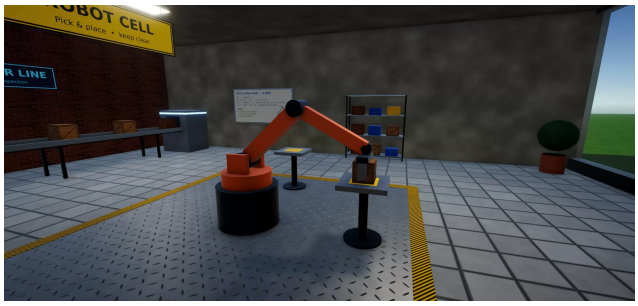
3 Environment Design: Materials, Colours, Textures and Lighting

Materials and colours. The project uses the Universal Render Pipeline (URP) with physically based *Lit* materials, so each surface has its own colour, metalness and smoothness. Steel parts are metallic and glossy; wood, fabric and concrete are matte. The palette is deliberately simple: neutral greys for the room, safety orange for the robot, yellow and black for hazard markings, and cyan accents on the signs and light strips. In total there are 46 materials.

Textures. I generated 12 tileable surface textures (floor tiles, concrete, brick, wood grain, brushed metal, diamond tread plate, hazard stripes, rubber belt, wooden crates, fabric, grass and leaves). Nine of them also have a matching normal map, which adds surface relief such as tile grout and brick mortar.

Each texture is tiled to match the real size of the surface it covers. A further six text textures (the signs, the whiteboard and the two monitor screens) were drawn with Python and Pillow. The glass panes are transparent and highly smooth, and they do not cast shadows, so daylight reaches the interior.

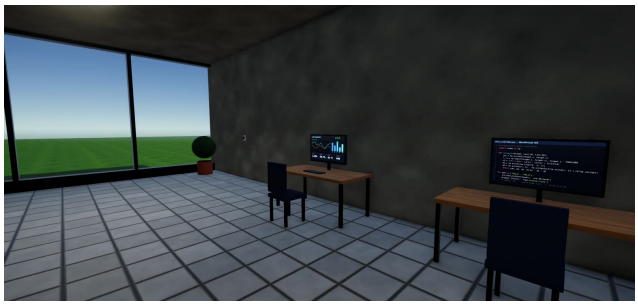
Lighting. A directional sun with soft shadows and a procedural sky light the scene, together with three-colour ambient light (sky, horizon and ground). Inside, six ceiling spot lights with soft shadows sit under glowing light panels. A reflection probe gives the metal surfaces realistic reflections, and light fog softens the horizon outside. Post-processing adds ACES tone mapping, bloom on bright surfaces, slight contrast and saturation adjustment, a vignette, ambient occlusion and anti-aliasing.



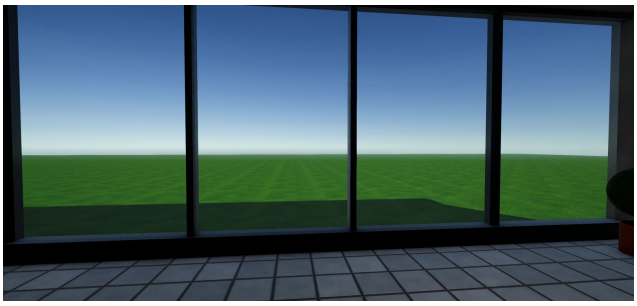
(a) Robot cell: the arm gripping a crate



(b) Conveyor line along the brick wall



(c) Workstations with monitors and chairs



(d) Glass wall overlooking the green ground

Figure 1: Zones of the lab, showing the materials, textures and lighting.

4 Camera Setup and Navigation

The user explores the lab with a **first-person camera** attached to a character controller (a capsule 1.8 m tall, 70° field of view). The character walks at 3.2 m/s, sprints at 6 m/s, can jump, and is affected by gravity. Walls, glass panes and furniture all have colliders, so the user cannot walk through them. Mouse look is lightly smoothed and the vertical angle is limited to ±80°. Pressing **V** switches to a **third-person camera** behind a simple avatar. This camera moves closer when a wall gets in the way, so it never clips through geometry. Pressing **T** starts a **guided tour**: the camera glides along a smooth path through eight viewpoints with cinematic black bars. This tour was used to record the submission video. An on-screen display shows the lab name, the current camera mode, a crosshair and a controls panel.

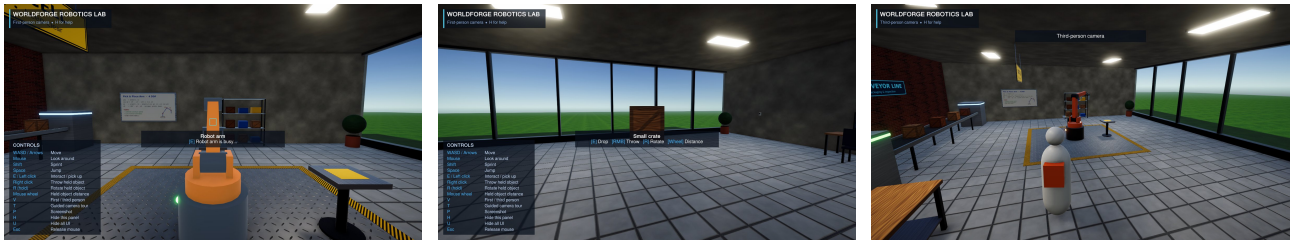
Key	Action	Key	Action
W A S D / arrows	Move	E / left click	Interact, pick up, drop
Mouse	Look around	Right click	Throw held object
Shift	Sprint	R (hold)	Rotate held object
Space	Jump	V	First / third person
T	Guided camera tour	H / U	Hide help / hide all UI

Table 2: Navigation and interaction controls.

5 Interaction

When the user aims at an interactive object, the object brightens slightly and a label below the crosshair names it and shows the available action. Pressing **E** or clicking performs that action:

- **Robot arm:** runs a full pick-and-place cycle. The arm moves above the crate, lowers, closes its gripper, lifts the crate, turns and places it on the other table, then returns home. Its joint angles are calculated in real time so the gripper always reaches the target, and a status lamp on the pedestal shows whether the arm is busy.
- **Conveyor belt:** stops or starts the line. The crates move along the belt while its surface scrolls.
- **Light switch:** turns the six ceiling lights and their glowing panels on or off.
- **Workbench items:** the crate, ball and toolbox are physics objects. The user can pick them up, carry and rotate them, drop them or throw them; they then fall, bounce and collide realistically.



(a) Aiming at the robot arm **(b)** Carrying a crate **(c)** Third-person camera

Figure 2: Interaction prompts, carrying an object, and the third-person view.

6 Tools Used

Tool	Purpose
Unity 6.3 LTS (6000.3.26f1), Unity Hub	Game engine and editor used to build, test and run the environment
Universal Render Pipeline 17.3	Physically based materials, real-time lighting, shadows and post-processing
Unity Input System 1.20	Keyboard and mouse input for navigation and interaction
Unity Recorder 5.1.7	Recording the guided tour at 2560×1440, 60 fps
C#	Scripts for movement, cameras, interactions and the guided tour
Python 3 with Pillow	Drawing the sign, whiteboard and monitor-screen textures
FFmpeg	Converting the recorded video to H.264 MP4
Git and GitHub, Ubuntu 24.04	Version control; development operating system

Table 3: Software used.

7 Development Process

1. **Planning.** I chose the robotics-lab theme, sketched a floor plan with five zones, and mapped each assignment requirement to concrete features.
2. **Project setup.** I created a Unity 6 project from the URP 3D template and configured the input system, quality settings and post-processing.
3. **Textures and materials.** I created the tileable surface textures and the text textures, then set up a material for each surface type with suitable colour, metalness and smoothness.
4. **Scene assembly.** I built the room shell first, then each zone from primitives at real-world scale. The robot arm was built as a chain of joints so each part moves with the one before it. The whole scene can be regenerated from the editor so it stays consistent.
5. **Lighting.** I added the sun, sky, ceiling lights, reflection probe and post-processing, then refined them by comparing screenshots. For example, I brightened the dark ceiling and toned down the object highlight.

6. **Navigation and interaction.** I added the first- and third-person controller, the interaction system, the robot arm's pick-and-place behaviour, the conveyor, the light switch, the objects that can be picked up, and the guided tour.
7. **Testing.** I play-tested in the editor and ran an automated check on a standalone build. The check confirmed the robot cycle, conveyor movement, light switching, walking, wall collisions, picking up and dropping objects, and the tour, and all checks passed.
8. **Simplification.** An earlier version also had an outdoor garden, a door and day/night cycling. For the submission I reduced the scope to a focused, polished lab with the glass wall and green ground.
9. **Recording.** I recorded the guided tour with Unity Recorder at QHD (2560×1440, 60 fps) and converted it to MP4.

8 Conclusion

The WorldForge Robotics Lab successfully demonstrates the creation of an interactive 3D virtual environment using Unity 6.3 LTS. The project integrates primitive-based 3D modelling, realistic materials and textures, lighting, and post-processing to create an immersive robotics laboratory. First-person and third-person navigation, collision detection, object manipulation, conveyor control, and the robot arm's pick-and-place mechanism provide a range of interactive experiences. The guided camera tour further showcases the environment and its key features. Overall, the project applies fundamental concepts of virtual environment development to build a functional, visually detailed, and interactive simulation.

9 References

1. Unity Technologies, *Unity 6 User Manual*. Available at: `https://docs.unity3d.com/Manual/`
2. Unity Technologies, *Universal Render Pipeline (URP) Documentation*. Available at: `https://docs.unity3d.com/6000.0/Documentation/Manual/urp/`
3. Unity Technologies, *Input System Documentation*. Available at: `https://docs.unity3d.com/Packages/com.unity.inputsystem@1.0/manual/`
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